

WASTEWISE-ML · DESIGN DECISIONS REGISTER

WasteWise-ml design decisions register

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DOCUMENT CONTROL

Document ID	WML-DEC-001	Version	0.4
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	MIT
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/WasteWise-ml		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-02	Amish Chadha	Approved follow-ups carried out; value engineering savings corrected (the reference phone is already under USD 150)
0.3	2026-10-02	Amish Chadha	Amish approved the recommendations for open items 1 to 6 on 2026-10-02 (WML-DDR-003 accepted, lens offset, no lamp, bench-clamped rig, SWaCH in Pune as first candidate partner, MIT-only contributing guide); moved to decisions made
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-09-30	Amish Chadha	Register opened with the build plan; open items from WML-DDR-001 to WML-DDR-003 and the review note

WasteWise-ml design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

None. All open decisions were decided on 2026-10-02.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The reference phone's width and thickness, the size of its camera bump and how far its lens is from the centre line across the width	They set the tray's inside width, the window and the arm's length; the model assumes 76 x 9 mm, a 30 x 30 mm bump and a centred lens	WML-DDR-003, A1
2	The phone's main camera is a 26 mm-equivalent lens, or the picture size at 440 mm is measured	The picture's fit on the mat (586 x 439 mm on 700 x 450 mm) depends on it	WML-CAL-001, section 6
3	The bench at the pilot site (first candidate: SWaCH, Pune) has a square back edge, a top 20 to 50 mm thick and room under it for a G-clamp's lower jaw	The board is clamped to that edge	WML-DDR-003, P1
4	The USB cable reaches about 0.93 m with a loop to spare and fits the phone's port	The cable runs up the stand	WML-DDR-003, P5

■ Value engineering

Value-engineering target: none set (a hypothetical control target is not defined here, because `budget_usd` is null in this software repository). Estimated cost of the constructable design: USD 309 for one station, up from USD 285 because the stand is now made and clamped to the bench (WML-DDR-003), and USD 1,348 for a two-station pilot with the dataset, evaluation set and compute. Main cost drivers and savings worth trying:

- The reference phone is the largest line (USD 130), then the seven bins (USD 70), the hazard box and platform scale (USD 25 each), the power bank and cable (USD 15) and the stand clamps and fixings (USD 14).
- The dataset and evaluation set labeling effort (USD 430 and USD 260) dominates the two-station pilot; its rate is set with the partner.
- Savings worth trying: the user's own phone, where it meets R12, in place of the USD 130 reference phone (the reference phone is already priced under the USD 150 ceiling of R12, so a cheaper phone is not counted as a saving). No lamp over the mat was decided on 2026-10-02 (option a), so the estimate already carries no lamp.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	D1 to D8: single item on a mat; MobileNetV3-Large with TensorFlow Lite and ONNX exports; keep the composite class and the unsure and hazard outputs; per-site grade file; tap-to-scan; consent, opt-in upload, no faces and co-ownership; labeling pay set with the partner at or above the local living wage; one reference phone per station for the evaluation	Amish: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."	WML-DDR-001
2026-09-25	N1: at least 400 hazard items in the evaluation set; N2: ZeroWaste excluded from training; N3: camera 440 mm above the mat; O3: kit option for MIT labels on drawing sheets (cross-repo action)	Amish: "i accept all your recommendations, go with them across all repos."	WML-DDR-002
2026-10-02	Design for construction of the scanning rig accepted: clamped base board, square tube post and arm with corner plates, foot angles, folded phone tray, cable route and mat placed by the board (P1 to P6), as made	Amish: "i approve your recommendations for all 555 open decisions."	WML-DDR-003, P1 to P6
2026-10-02	Phone lens off the centre line: option (a), lay the phone with its lens on the side away from the post and cut the arm shorter by the measured lens offset, only after the reference phone is bought and measured	Amish: "i approve your recommendations for all 555 open decisions."	WML-DDR-003, A1
2026-10-02	Lighting: option (a) for the prototype, no lamp; the site light is recorded with every photo (a light reading or a gray card in a corner of the frame); a lamp is decided only after the first site visit	Amish: "i approve your recommendations for all 555 open decisions."	WML-DDR-003, A2
2026-10-02	Sites without a suitable bench: option (a) for the prototype, the bench-clamped rig, with handheld use at sites without a square-edged bench about 600 mm deep	Amish: "i approve your recommendations for all 555 open decisions."	WML-DDR-003, A3
2026-10-02	First partner, city and buyers: a member-owned waste picker cooperative that already sorts dry waste for scrap buyers; first candidate to approach SWaCH in Pune, India, with its existing scrap dealers as the buyers and Pune as the city	Amish: "i approve your recommendations for all 555 open decisions."	WML-DDR-001, O1
2026-10-02	Contributing guide: remove the CERN-OHL-S sentence from CONTRIBUTING.md and state that all contributions are licensed under MIT	Amish: "i approve your recommendations for all 555 open decisions."	WML-DDR-001, O2